Tamal Joyti Roy

PhD Candidate | Measurement, Quantitative Methods & Learning Sciences
University of Houston, College of Education

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Education

Ph.D. in Measurement, Quantitative Methods & Learning Sciences 2023–Present
University of Houston, Houston, TX
Focus: Quantitative Methods

Bachelor of Science in Computer Science and Engineering 2016–2020
Northwestern University, Khulna, Bangladesh
Focus: Data Mining

Professional Experience

Research Assistant June 2026– Present
Tilman J. Fertitta Family College of Medicine, University of Houston | Mentor: Dr. Ben King

- Manage and document quantitative datasets across multiple research projects, including data cleaning, coding, variable construction, and quality assurance.
- Conduct reliability, dimensionality, and preliminary psychometric analyses for pilot measures, including medical respite fidelity and medical student experience instruments.
- Support measure development and validation through statistical analysis, interpretation, and preparation of methods, results summaries, abstracts, manuscripts, and reports.
- Maintain identifiable and analytic data in accordance with federal, state, and institutional requirements.

Summer Teaching Assistant June 2026–July 2026
College of Education, University of Houston | Mentor: Dr. Kelli Drenner

- Supported two undergraduate courses: *Child and Adolescent Health* and *Health Aspects of Human Sexuality*.
- Graded assignments, quizzes, and discussions; provided feedback and monitored student progress in Canvas LMS.
- Assisted with student communication, course administration, and grade-record maintenance.

Teaching Assistant January 2026–May 2026
College of Education, University of Houston | Mentor: Dr. Ken Ripperger-Suhler

- Evaluated weekly assignments, quizzes, literature reviews, and deliverables for four semester-long group projects.
- Provided individualized feedback on health behavior theory and evidence-based intervention design.
- Supported Canvas course management, deadline tracking, participation monitoring, and academic-integrity procedures.

Research Assistant – BOUNCE June 2024–Aug 2026
College of Education, University of Houston | NSF AISL Grant | PI/Mentor: Dr. Norma Olvera

- Manage data collection, cleaning, and analysis for multiple health intervention programs, including the Diabetes Prevention Program and Nutrition Ambassador Program.
- Conduct quantitative analyses and contribute to manuscripts, reports, and conference products.
- Mentor graduate and undergraduate interns in quantitative research methods, data management, and research dissemination.

Graduate Assistant July 2025–September 2025

C.T. Bauer College of Business & Tilman J. Fertitta Family College of Medicine, University of Houston | Mentor: Dr. M. Mahbub Hossain

- Conducted systematic and scoping review work on first-generation college student success, grit, artificial intelligence, and self-efficacy in health professions education.
- Analyzed HINTS 7 data on digital health and wearable technology use and contributed to scholarly manuscripts and commentaries.
- Synthesized evidence and performed statistical analyses across multiple concurrent research projects.

Instructional Assistant June 2024–August 2024

C.T. Bauer College of Business, University of Houston | Mentor: Dr. M. Mahbub Hossain

- Conducted statistical analyses for a funded research project and translated findings into project deliverables.
- Contributed to manuscript preparation and research synthesis.

Teaching Assistant June 2024–August 2024

College of Education, University of Houston | Mentor: Dr. Toya Conston

- Supported two asynchronous course sections of approximately 50 students each through office hours, individualized assistance, and Canvas administration.
- Maintained course materials and responded to student questions about psychological concepts and course requirements.

Research Assistant August 2023–May 2024

College of Education, University of Houston | Mentor: Dr. Tiffany J. Davis

- Conducted statistical analyses of student-success outcomes and developed reproducible data-preprocessing workflows.
- Created analytical reports and data visualizations using Tableau and Power BI.

Data Scientist November 2021–June 2023

United We Care, India | Full-time

- Developed predictive models and data-science workflows using PyTorch and scikit-learn to support business decision-making.
- Applied statistical analysis and data visualization to translate complex data into stakeholder-facing reports.
- Supported generative-AI and SEO analytics initiatives through data-driven experimentation and implementation.

Peer-Reviewed Publications

Journal Articles

1. **Roy, T. J.**, & Bonya, B. M. (2026). Social isolation, eHealth engagement, and psychological distress among U.S. adults. *Health Education & Behavior*. https://doi.org/10.1177/10901981261479302
2. **Roy, T. J.**, Saha, N., Chavez, S., Liaw, W., & Hossain, M. M. (2026). Grit in undergraduate medical education: A scoping review. *Medical Science Educator*. Advance online publication. https://doi.org/10.1007/s40670-026-02799-0
3. Olvera, N., Matthews-Ewald, M. R., **Roy, T. J.**, Scherer, R., Garcini, L. M., & Arbona, C. (2026). Associations among anxiety, sleep quality, and binge eating in Hispanic and African American/Black early adolescents. *Children, 13*(6), 761. https://doi.org/10.3390/children13060761
4. Saha, N., **Roy, T. J.**, & Hossain, M. M. (2026). Preventing health information disorders (HIDs) in college students: Perspectives on health information hygiene in higher education. *Journal of American College Health*, 1–5. https://doi.org/10.1080/07448481.2026.2630052
5. Huynh, A. L., **Roy, T. J.**, Jackson, K. N., Lee, A. G., Liaw, W., & Hossain, M. M. (2025). Applications of artificial intelligence-based conversational agents in healthcare: A systematic umbrella review. *International Journal of Medical Informatics*, 106204. https://doi.org/10.1016/j.ijmedinf.2025.106204
6. Olvera, N., Scherer, R., **Roy, T. J.**, Chavez Cubas, N. W., Matthews-Ewald, M. R., & Arbona, C. (2025). Visual voices: Hispanic and African American youth perspectives on neighborhood and school health. *Children, 12*(9), 1165. https://doi.org/10.3390/children12091165
7. Olvera, N., Scherer, R., Wu, W., **Roy, T. J.**, Matthews-Ewald, M. R., Fan, W., & Arbona, C. (2025). The role of doctor visits, body image discrepancy, and perceived health in predicting medical weight problem diagnosis. *Healthcare, 13*(17), 2135. https://doi.org/10.3390/healthcare13172135
8. Olvera, N., **Roy, T. J.**, Scherer, R., Matthews-Ewald, M., Fan, W., & Arbona, C. (2025). Emotional eating in Hispanic youth: The role of anxiety and sleep quality. *Nutrients, 17*(9), 1588. https://doi.org/10.3390/nu17091588
9. Hossain, M. M., Saha, N., Rodela, T. T., Tasnim, S., Nuzhath, T., **Roy, T. J.**, Burdine, J. N., Ahmed, H. U., McKyer, E. L. J., Basu, B. K., & Ma, P. (2022). Global research on syndemics: A meta-knowledge analysis (2001–2020). *F1000Research, 11*, 253. https://doi.org/10.12688/f1000research.74190.2

Manuscripts Under Review / In Revision

1. Understanding medical weight problem diagnosis in Hispanic women and men: The role of healthcare utilization, body image, and perceived health status.
2. Self-efficacy in digital nursing education: A scoping review.

Manuscripts Under Preparation

1. The role of self-efficacy and regulation difficulties in predicting academic procrastination.
2. The role of anxiety, body discrepancy, and sleep on body mass index among Latino youth.
3. The impact of teacher and student beliefs on math and science engagement.
4. Linking growth mindset to academic engagement through college belongingness and self-efficacy.
5. Academic belonging in the age of artificial intelligence: A scoping review.
6. Predicting online health information seeking among U.S. adults: A machine learning classification analysis of HINTS 7 with implications for digital health equity.
7. From Growth Mindset to Academic Engagement: The Mediating Roles of College Belonging and Self-Efficacy.

Book Chapters

1. Shelton, L., Nguyen, K., Joseph, T., **Roy, T. J.**, Roy, P., Alarcón, J., & Olvera, N. (in press). How is our community healthy? Program design and lessons learned while training college students to facilitate photovoice projects with children. In *The Palgrave Handbook on Participatory Action Research with Children*. Palgrave Macmillan.
2. Roy, D., Mahmood, M. A., & **Roy, T. J.** (2023). An efficient approach to assess the soil quality of Sundarbans utilizing hierarchical clustering. In *Applied Intelligence for Industry 4.0* (pp. 79–89). Chapman and Hall/CRC. DOI
3. **Roy, T. J.**, & Mahmood, M. A. (2023). Global warming and Bangladesh: A machine learning approach to analyze the warming rate utilizing neural network. In *Lecture Notes in Electrical Engineering* (pp. 19–30). Springer Nature Singapore.

Conference Presentations

1. **Roy, T. J.**, Saha, N., Choi, C., Roy, P., & Master, A. (2026, August 6–8). Fostering sense of belonging in science education: The role of self-concept, interest, and students' perceptions of teachers' beliefs [Poster]. APA Annual Convention, Washington, D.C.
2. Saha, N., Roy, P., **Roy, T. J.**, Chavez Cubas, N. W., Scherer, R., & Olvera, N. (2026, April). A multilevel analysis of socioeconomic status disparities: The role of school context in shaping student outcomes [Roundtable]. AERA Annual Meeting, Los Angeles, CA.
3. **Roy, T. J.**, Das, J., & Hossain, M. M. (2025, November 2–5). Factors associated with online medical record use among U.S. adults with hypertension [Presentation]. APHA Annual Meeting, Washington, DC.
4. **Roy, T. J.**, Das, J., & Hossain, M. M. (2025, November 2–5). Predictors of health and wellness apps use among smartphone users in the U.S. [Presentation]. APHA Annual Meeting, Washington, DC.
5. **Roy, T. J.**, Das, J., & Hossain, M. M. (2025, November 2–5). Prevalence and correlates of wearable health technology use among people with diabetes in the U.S. [Presentation]. APHA Annual Meeting, Washington, DC.
6. Das, J., **Roy, T. J.**, & Hossain, M. M. (2025, November 2–5). Association between online health information-seeking behavior and perceived health-related misinformation among older adults in the U.S. [Presentation]. APHA Annual Meeting, Washington, DC.
7. Saha, N., Guzman, Y. G., **Roy, T. J.**, Chavez Cubas, N. W., Daniel, P., Nguyen, K., & Olvera, N. (2025, November 2–5). Association between online health information-seeking behavior and perceived health-related misinformation among older adults in the U.S. [Presentation]. APHA Annual Meeting, Washington, DC.
8. **Roy, T. J.**, Fan, W., Olvera, N., & Arbona, C. (2025, August). The role of self-efficacy and regulation difficulties in predicting academic procrastination [Poster]. APA Annual Convention, Denver, CO.
9. **Roy, T. J.**, Saha, N., Nguyen, N., Scherer, R., Fan, W., Arbona, C., & Olvera, N. (2025, August). The role of anxiety, body discrepancy, and sleep on body mass index among Latino youth [Poster]. APA Annual Convention, Denver, CO.
10. **Roy, T. J.**, Roy, P., Scherer, R., Fan, W., Arbona, C., & Olvera, N. (2025, August). Emotional eating in Hispanic adolescents: Role of anxiety and sleep quality [Poster]. APA Annual Convention, Denver, CO.
11. Huynh, A. L., **Roy, T. J.**, Jackson, K. N., Lee, A., & Hossain, M. M. (2024). Effectiveness of AI-based chatbots in healthcare: An umbrella review. International Conference on Information Systems (ICIS), TREOS.
12. Ashiq Mahmood, M., **Roy, T. J.**, Ashiqul Amin, M., Roy, D., Mohanta, A., Fayez Dipty, F., & Mitra, S. (2023). A hybrid approach to find COVID-19 related lung infection utilizing 2-bit image processing. In *Lecture Notes in Networks and Systems* (pp. 119–127). Springer Nature Singapore.
13. **Roy, T. J.**, Mahmood, M. A., & Mohanta, A. (2022). An efficient approach to validate COVID-19 related vaccine myths utilizing LDA algorithm. *2nd International Conference on Computing Advancements*, 53–58.

14. **Roy, T. J.**, Mahmood, M. A., Mohanta, A., Roy, D., Jyoti, J. T., & Ghosh, P. K. (2022). A machine learning approach to analyze the performance of Bangladesh cricket in T20. *2022 International Conference on Innovations in Science, Engineering and Technology (ICISET)*, 129–134.
15. Roy, D., **Roy, T. J.**, Mahmud, I., & Alvi, N. (2022). An efficient approach to predict fear of human's mind during COVID-19 outbreaks utilizing data mining technique. In *Advances in Intelligent Systems and Computing* (pp. 41–51). Springer Singapore.
16. **Roy, T. J.**, Mahmood, M. A., & Roy, D. (2021). A machine learning model to predict earthquake utilizing neural network. *2021 International Conference on Computer, Communication, Chemical, Materials and Electronic Engineering (IC4ME2)*, 1–4.
17. **Roy, T. J.**, Mahmood, M. A., & Roy, D. (2021). An efficient approach to identify key factors of failure of Bangladesh cricket team in test cricket utilizing hypothesis testing and clustering. *2021 5th International Conference on Electrical Information and Communication Technology (EICT)*, 1–6.
18. **Roy, T. J.**, Mahmood, M. A., Mohanta, A., & Roy, D. (2021). An analytical approach to predict the COVID-19 death rate in Bangladesh utilizing multiple regression and SEIR model. *2021 IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, 42–45.
19. Roy, D., Mahmood, M. A., **Roy, T. J.**, & Mohanta, A. (2021). A two-step clustering approach to measure soil quality of Sundarban based on organic carbon and bulk density. *2021 IEEE RAAICON*, 85–88.

Selected Research Projects

Latent Profiles of Personality and Psychopathology in Voice Disorders October 2026–Present
Department of Communication Sciences & Disorders, University of Houston | PI/Mentor: Dr. Brett Welch | Role: Data Analyst

- Conduct data management and confirmatory factor analyses in R (lavaan) to derive factor scores for personality aspects (BFAS), psychopathology, depression (CES-D), anxiety, and perceived stress.
- Apply finite mixture modeling (latent profile analysis) to identify psychological subgroups among adults with muscle tension dysphonia, phonotraumatic vocal hyperfunction, and vocally healthy controls.

Motivation, Grit, and Academic Engagement September 2024–Present
University of Houston, PHLS | IRB #STUDY00004949 | Role: Co-PI

- Designed an IRB-approved study examining motivation, grit, and academic engagement among college students; oversee quantitative analysis and dissemination.

Systematic Review: First-Generation College Student Success June 2024–Present
University of Houston

- Developed systematic search and screening procedures and synthesized evidence on academic success factors among first-generation college students in the United States.

CoderKids: Impact of Coding Programs on Children's Learning April 2024
University of Houston, PHLS | IRB #STUDY00004640 | Role: Graduate Student Researcher

- Contributed to an IRB-approved study investigating the relationship between coding-program participation, engagement, and learning outcomes among children.

Awards, Scholarships, and Recognition

- PHLS Doctoral Student First Author Award (Fall 2026) | $100.
- PHLS Doctoral Student First Author Award (Spring 2026) | $100.
- Basheer A. Hashmi Scholarship, 2026–2027 | $5,000.
- Graduate Student Poster Award Finalist, APA Division 15, Top 10 of 619 submissions, 2025.
- Joshua Weinstein Memorial Endowed Scholarship, University of Houston, 2025–2026 | $2,000.
- PHLS Travel Fellowship, University of Houston, 2025–2026 | $500.
- TA & RA Waiver (54.212 PH), University of Houston, 2025–2026 | $9,180.
- TA & RA Waiver (54.212 PH), University of Houston, 2024–2025 | $9,180.
- Graduate Tuition Fellowship, University of Houston, 2024–2025 | $8,162.
- Competitive SCH WCR (54.213 PH), University of Houston, 2023–2024 | $6,240.
- TA & RA Waiver (54.212 PH), University of Houston, 2023–2024 | $4,680.
- Graduate Tuition Fellowship, University of Houston, 2023–2024 | $9,929.

Academic Service

Ad Hoc Peer Reviewer

- *Cureus Journal of Medical Science* (2023–Present)
- *JBI Evidence Synthesis* (2025–Present)
- *Frontiers in Psychology – Educational Psychology* (2026–Present)
- *Health Promotion Practice* (2026–Present)
- *Frontiers in Public Health* (2026–Present)
- *Informatics for Health and Social Care* (2026–Present)

Professional Memberships

- American Educational Research Association (AERA), Student Member (2023–2024)
- American Public Health Association (APHA), Student Member (2025–Present)
- American Psychological Association (APA), Student Member (2025–Present)

Technical Skills & Certifications

Technical Skills

- **Statistical & Quantitative Methods:** Mplus 8.10 (SEM, mediation, BCa bootstrapping, WLSMV), HLM (multilevel modeling), SPSS, R (lavaan, replication-weight methods).
- **Programming & Machine Learning:** Python (scikit-learn, PyTorch, TensorFlow, OpenCV), R, C, C++, Java.
- **Data Visualization:** Tableau, Power BI, Matplotlib, RapidMiner Studio.
- **Research & Workflow Tools:** Excel, LaTeX, Git, Canvas LMS.

Certifications

- Social-Behavioral-Educational Researchers, CITI Program, 2023–2026.
- AI Productivity Hacks, LinkedIn Learning, 2024.
- Intermediate Python for Non-Programmers, LinkedIn Learning, 2024.
- Introduction to Prompt Engineering for Generative AI, LinkedIn Learning, 2024.

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